

Adversarial Example Does Good: Preventing Painting Imitation from Diffusion Models via Adversarial Examples

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- We highlight how diffusion models advance images, video, and 3D.
- We note commercial tools enable high-fidelity, controllable creation.
- We focus on textual inversion and image-to-image as high-risk vectors.

Specific Problem: Style Imitation and Copyright Risk

- We observe unauthorized scraping enables style cloning without consent.
- We see pseudo-words from few images make imitation easy.
- We aim to provide a practical, technical protection for creators.

Key Challenges Blocking Protection

- We find diffusion adversarial examples underdefined vs. classifier attacks.
- We cannot compute image likelihood under a diffusion model directly.
- We must keep perturbations human-similar while breaking conditioning.

Core Idea: Adversarial Examples that Block Feature Extraction in DMs

- We define adversarial examples as inputs that reduce usable signals.
- We align the attack with diffusion training to lower model affinity.
- We protect images so DMs cannot extract style or content cues.

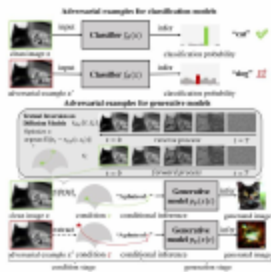


Figure: 分类与扩散模型对抗样本流程对比：在扩散模型中诱导OOD，阻碍条件特征提取；示例为文本反演（Gal'22）。

Method Foundation: Objective and Threat Model

- We state a single goal: minimize the model's likelihood of the perturbed image within a small, human-similar change.
$$x' = \arg \min_{x' : \|x' - x\|_\infty \leq \epsilon} p_\theta(x')$$
- We cannot compute likelihood directly, so we optimize a training-aligned surrogate.
- We assume white-box knowledge for crafting; we test transfer to black-box systems.

Contribution Overview

- We present a principled framework for adversarial examples in diffusion.
- We propose AdvDM, a Monte Carlo, training-aligned gradient method.
- We validate broadly across datasets, tasks, and commercial systems.
- We analyze strengths, limits, and partial defenses.

AdvDM Overview: Monte Carlo Ascent on the Training Loss

- We randomly select a denoising timestep each iteration.
- We compute the diffusion training loss residual at that step.
- We backpropagate to the input and step to increase this loss.
- We project back to the allowed perturbation range after each step.
- We repeat to produce a protected image that resists conditioning.
- We outperform classifier-PGD and naive LDM-PGD baselines.

- We use the encoder and denoiser to expose informative latents.
- We, per step, draw a reverse-process latent path and a timestep.
- We backpropagate through encoder and denoiser to input pixels.
- We use small per-step changes and clip to the allowed range.

- We let gradients flow encoder → U-Net denoiser → input image.
- We handle conditioning text/controls consistently during backprop.
- We batch multiple latent draws to stabilize gradient estimates.
- We use mixed precision where safe to cut memory/runtime.
- We cache encodings when possible to reduce overhead.

What Do Protected Images Look Like?

- We keep perturbations subtle and human-similar.
- We observe DMs fail to extract style/content despite similarity.
- We show examples across categories: cat, sheep, airplane.



Figure: LSUN猫/羊/飞机：上为干净样本，下为AdvDM对抗样本

Experimental Setup: Datasets, Scenarios, and Metrics

- We evaluate on LSUN-cat/sheep/airplane and WikiArt.
- We test text-to-image via textual inversion and image-to-image.
- We report FID and Precision; we follow Kynkäänniemi et al. (2019).
- We include Stable Diffusion and commercial platforms.

- We observe sharp FID↑ and Precision↓ across LSUN.
- We confirm the model cannot learn usable pseudo-words (S^*).
- We evaluate with FID and Precision (Kynkäänniemi et al., 2019).

Table: Text-to-image with textual inversion on LSUN (higher is better for FID↑, lower is better for Precision↓).

Method	LSUN-CAT			LSUN-SHEEP			LSUN-AIRPLANE		
	FID↑	Prec.↓	Recall	FID↑	Prec.↓	Recall	FID↑	Prec.↓	Recall
No Attack	34.94	0.5643	0.1531	32.81	0.6378	0.1228	39.22	0.5016	0.2765
AdvDM	127.04	0.1708	0.0610	203.50	0.0058	0.3780	169.67	0.0263	0.3235

- We find pseudo-words from protected sets fail to reproduce concepts.
- We see target attributes lost versus clean-trained S^* .
- We note qualitative degradation matches FID and Precision.



Figure: 基于文本反演的文到图生成：左为干净，右为对抗

- We show protected sets break brushwork and palette capture.
- We observe the model cannot recover composition cues.
- We see the effect persist across image-to-image strengths.

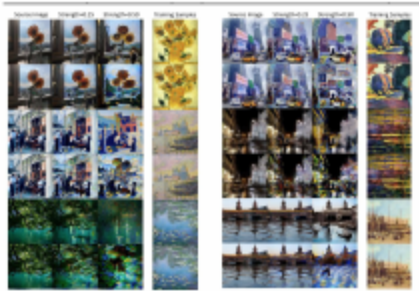


Figure: WikiArt风格迁移对比：文本反演提取风格；上行为干净样本，下为对抗样本；LDM对抗样本失效。

- We cannot compute diffusion likelihoods in closed form.
- We instead maximize the standard diffusion training loss.
- We tie optimization to reverse-process sampling steps.
- We use randomness to probe many denoising stages.
- We find this surrogate reliably reduces model affinity.

- We randomly pick a denoising timestep each update.
- We sample a latent trajectory from the reverse process.
- We reuse multiple latent draws to stabilize gradients.
- We average per-draw losses before taking an update.
- We keep samples per update modest for speed.

- We take signed gradient ascent to increase training loss.
- We project back to the allowed perturbation range each step.
- We clip gradients to curb exploding updates.
- We add light randomization to avoid overfitting a timestep.
- We tune step sizes to balance speed and stability.

- We enforce an l-infinity budget to bound pixel changes.
- We schedule step sizes to ensure steady progress.
- We target perceptual imperceptibility on clean monitors.
- We monitor image statistics to avoid visible artifacts.

- Initialize the protected image as the original.
- Repeat for a fixed number of iterations:
 - — Randomly choose a denoising timestep.
 - — Sample one or more reverse-process latent paths.
 - — Compute the diffusion training loss and its gradient.
 - — Take a small ascent step and project to the budget.
- Return the protected image for publishing.

- We craft with white-box access to an aligned diffusion model.
- We test transfer to closed systems without internals.
- We expect transfer because training signals are shared.
- We observe strong black-box effects in commercial tools.

- We spend most cost in encoder and denoiser passes.
- We parallelize latent samples within a minibatch.
- We trade more samples for stronger protection.
- We use mixed precision and caching to cut runtime.

- We see more samples strengthen protection but cost time.
- We advise choosing steps by desired strength and budget.

Table: AdvDM under different sampling steps. Bold indicates strongest protection per column (by arrows).

Sampling Steps	FID↑	Prec.↓	Inference Time
10	122.9	0.05	1.803
40	186.05	0.037	6.342
1000	211.88	0.011	166.6

- We suggest 10 steps for fast, moderate protection.
- We use 40 steps for strong default protection.
- We reserve 1000 steps for maximal strength studies.
- We align choices with the ablation table trade-offs.

Ablation: Perturbation Budget (Table)

- We see larger budgets strengthen protection consistently.
- We note even modest budgets cause large degradation.
- We recommend balancing visibility and protection needs.

Table: Effects of AdvDM under different perturbation budgets on LSUN-Airplane.

Setting	FID↑	Prec.↓	Recall
No Attack	54.03	0.659	0.242
2/255	54.49	0.295	0.276
4/255	116.79	0.090	0.342
8/255	186.05	0.037	0.464

Transfer to Commercial Platforms

- We demonstrate black-box transfer to scenario.gg.
- We observe artifacts and lost stylistic coherence.
- We confirm breadth across artist styles and prompts.



Figure: 上: 基于马蒂斯画作的干净样本生成; 下: 对抗样本生成, 纹理混乱 (scenario.gg)

Choosing the Perturbation Budget

- We use 4–8/255 for strong, low-visibility protection.
- We push 16–32/255 for maximum robustness when allowed.
- We inspect images at native resolution to confirm quality.

Limitations and Practical Considerations

- We trade runtime for strength via sampling steps.
- We risk visibility at very large budgets.
- We see transfer vary across platforms and pipelines.
- We view defenses as open problems with partial mitigations.

- We introduce a principled diffusion adversarial framework.
- We provide AdvDM, a simple, effective, training-aligned method.
- We show robust, transferable protection across tasks and systems.
- We offer a practical tool that complements creator safeguards.

- We will strengthen black-box transfer and consent-aware conditioning.
- We will explore adaptive defenses and watermark integration.
- We will extend to video and 3D diffusion backbones.
- We aim for responsible AI-for-Art with creator protections.

- PGD adversarial training/attack: Madry et al., ICLR 2018.
- Stable Diffusion / Latent Diffusion Models: Rombach et al., CVPR 2022.
- Textual Inversion: Gal et al., 2022.
- DreamBooth: Ruiz et al., CVPR 2023.
- DiffPure: Nie et al., NeurIPS 2022.
- Precision/Recall for generative models: Kynkäänniemi et al., NeurIPS 2019.
- JPEG: Wallace, Comm. ACM, 1992. TV denoising: Rudin–Osher–Fatemi, Physica D, 1992.
- Super-Resolution (e.g., ESRGAN): Wang et al., ECCVW 2018.

- We appreciate your attention!
- We welcome questions and feedback.

- We thank collaborators, supporters, and the open-source community.
- Code: <https://github.com/mist-project/mist.git>